

Problem 1

- If n is odd
 - Consider x^i for $0 < i < n$ s.t. x is a rotation

$$x^j x^i x^{-j} \text{ for } 0 \leq j < n$$

$$= x^i$$

$$yx^j x^i yx^j \text{ for } 0 \leq j < n$$

$$= yyx^{n-j-i} x^j = x^n x^{-i} x^{j-j} = x^{-i}$$
 Thus $C_{x^i} = \{x^i, x^{-i}\} = C_{x^{-i}}$
 - Consider y s.t. y is a reflection

$$x^j y x^{-j} \text{ for } 0 \leq j < n$$

$$= y x^{n-j} x^{-j} = y x^{-2j}$$
 Since n is odd we can solve $k \equiv -2j \pmod{n}$
 thus $C_y = \{yx^k \mid k \in \mathbb{Z}\} = \{yx^k \mid k \in \{0, 1, 2, \dots, n-1\}\}$
 Thus $|D_n| = \sum_{[x^i]} |C_{x^i}| + |C_y| = |\{e\}| + \sum_{[x^i] \setminus [e]} |C_{x^i}| + |C_y| = 1 + (n-1) + n = 2n$
- If n is even
 - It is the same for rotations
 - Consider y s.t. y is a reflection

$$x^j y x^{-j} \text{ for } 0 \leq j < n$$

$$= y x^{n-j} x^{-j} = y x^{-2j}$$
 thus $C_y = \{y, yx^2, yx^4, \dots, yx^{n/2-1}\}$
 - Consider yx

$$x^j y x x^{-j} \text{ for } 0 \leq j < n$$

$$= y x^{n-j} x^{1-j} = y x^{1-2j}$$
 thus $C_{yx} = \{yx, yx^3, yx^5, \dots, yx^{n-1}\}$
 Thus $|D_n| = \sum_{[x^i]} |C_{x^i}| + |C_y| + |C_{yx}|$

$$= |\{e\}| + |\{x^{n/2}\}| + \sum_{[x^i] \setminus [e] \cup [x^{n/2}]} |C_{x^i}| + |C_y| + |C_{yx}| = 1 + 1 + 2(n/2 - 1) + 2(n/2) = 2n$$

Problem 2

- Conjugacy class of 1:

$$\{1\}$$

Since 1 is the identity element, it commutes with all other elements:

$$h \cdot 1 \cdot h^{-1} = 1 \quad \text{for all } h \in Q_8.$$

- Conjugacy class of -1 :

$$\{-1\}$$

Similar to 1, the element -1 commutes with all other elements:

$$h \cdot (-1) \cdot h^{-1} = -1 \quad \text{for all } h \in Q_8.$$

- Conjugacy class of i and $-i$:

The conjugates of i by other elements of Q_8 are:

$$1 \cdot i \cdot 1^{-1} = i, \quad (-1) \cdot i \cdot (-1)^{-1} = i,$$

$$j \cdot i \cdot j^{-1} = -i, \quad k \cdot i \cdot k^{-1} = -i.$$

Therefore, the conjugacy class of i is:

$$\{i, -i\}.$$

Similarly, for $-i$:

$$j \cdot (-i) \cdot j^{-1} = i, \quad k \cdot (-i) \cdot k^{-1} = i,$$

so i and $-i$ share the same conjugacy class.

- Conjugacy class of j and $-j$:

The conjugates of j are computed as follows:

$$1 \cdot j \cdot 1^{-1} = j, \quad (-1) \cdot j \cdot (-1)^{-1} = j,$$

$$i \cdot j \cdot i^{-1} = -j, \quad k \cdot j \cdot k^{-1} = -j.$$

Hence, the conjugacy class of j is:

$$\{j, -j\}.$$

Similarly, for $-j$:

$$i \cdot (-j) \cdot i^{-1} = j, \quad k \cdot (-j) \cdot k^{-1} = j,$$

so j and $-j$ are in the same conjugacy class.

- Conjugacy class of k and $-k$:

The conjugates of k are:

$$1 \cdot k \cdot 1^{-1} = k, \quad (-1) \cdot k \cdot (-1)^{-1} = k,$$

$$i \cdot k \cdot i^{-1} = -k, \quad j \cdot k \cdot j^{-1} = -k.$$

Therefore, the conjugacy class of k is:

$$\{k, -k\}.$$

Similarly, for $-k$:

$$i \cdot (-k) \cdot i^{-1} = k, \quad j \cdot (-k) \cdot j^{-1} = k,$$

so k and $-k$ share the same conjugacy class.

Thus, the conjugacy classes of Q_8 are:

$$\{1\}, \quad \{-1\}, \quad \{i, -i\}, \quad \{j, -j\}, \quad \{k, -k\}.$$

The sizes of these conjugacy classes are:

$$|\{1\}| = 1, \quad |\{-1\}| = 1, \quad |\{i, -i\}| = 2, \quad |\{j, -j\}| = 2, \quad |\{k, -k\}| = 2.$$

Therefore, the class equation of Q_8 is:

$$8 = 1 + 1 + 2 + 2 + 2.$$

Problem 3

- (a) • (i. $\implies$ iii.) Assume i.

Let $\varphi : G \rightarrow K$ s.t. $\varphi(g) = \varphi(hk) = k$

Well Defined: Let $g \in G$ by i. since $g = hk$ is unique this implies every input has one output.

Kernel: Let $g \in \ker(\varphi)$

$\varphi(g) = \varphi(hk) = e \implies k = e$ thus $g = h \in H$

Let $h \in H$ thus $he \in HK$ thus $\varphi(h) = \varphi(he) = e$ thus $H = \ker(\varphi)$

Surjective: Let $k \in K$ consider: $ek \in G$

$\varphi(ek) = k$ thus φ is surjective.

By the first isomorphism theorem this implies $K \cong G/H$

- (iii. $\implies$ ii.) Assume iii. with the map φ

Since φ is injective $\ker(\varphi) = \{e\} \implies \ker(\varphi) \subseteq K \cap H$ Let $l \in K \cap H \implies l \in K$ and $l \in H$

thus $\varphi(l) = lH$ but since $l \in H \implies \varphi(l) = H$ thus $l \in \ker(\varphi)$ thus $\ker(\varphi) = K \cap H = \{e\}$

Since φ is surjective this means for all $kH \in G/H$ there exists a $k' \in K$ s.t. $\varphi(k') = kH$

and since G/H partitions G this implies for all $g \in G \implies g \in jH$ for some $j \in K \implies$

$g = jh$ for some $h \in H$

thus $G = HK$

- (ii. $\implies$ i.) Assume ii.

Let $g \in G$ s.t. $g = hk = ij$ for $h, i \in H$ and $k, j \in K$

thus $i^{-1}h = jk^{-1} \implies$ that is in $H \cap K$ thus $i^{-1}h = e \implies i = h$ and $jk^{-1} = e \implies j = k$

thus all $g \in G$ are a unique product of an $h \in H$ and $k \in K$

- (b)